

Sensor Validation and Analysis

Drift Analysis – Temperature Sensors

To assess long-term stability, we analyzed drift in temperature readings from the DS18B20 and BME680 sensors using a stable portion of the 16-hour atmospheric logging data.

A linear trendline was fitted to the selected stable region for both sensors. The slope of the line represents the rate of drift over time.

Results:

DS18B20: Slope = $0.00001\text{ }^{\circ}\text{C/sec} \rightarrow 0.036\text{ }^{\circ}\text{C/hour}$

BME680: Slope = $0.000005\text{ }^{\circ}\text{C/sec} \rightarrow 0.018\text{ }^{\circ}\text{C/hour}$

Both sensors showed minimal temperature drift, with values well under 0.05°C per hour, indicating excellent stability for long-duration applications.

Drift Analysis - Humidity Sensors

Linear trendlines were fitted to the stable portion of the salt slurry test data. The slopes obtained were:

DHT22: $0.00002\text{ \%RH/sec} \rightarrow 0.072\text{ \%RH/hour}$

BME680: $0.00001\text{ \%RH/sec} \rightarrow 0.036\text{ \%RH/hour}$

Both sensors demonstrated good stability, with very low drift during the test period, making them suitable for long-duration monitoring tasks.

Accuracy and Offset Analysis

The mean offsets between sensor readings and the BME680 reference sensor were calculated:

DHT22 Humidity: $+2.27\text{ \%RH}$ (DHT22 reads higher)

DHT22 Temperature: $-0.53\text{ }^{\circ}\text{C}$ (DHT22 reads slightly lower)

DS18B20 Temperature: $-0.75\text{ }^{\circ}\text{C}$ (DS18B20 reads moderately lower)

These offsets are consistent and within acceptable ranges for low-cost sensors. Calibration can further improve accuracy.

Repeatability Test

To evaluate sensor repeatability, we conducted the same experiment under identical conditions twice and compared the readings.

Test Environment: Controlled humidity (Salt slurry at ~75% RH)

Sensor Evaluated: DHT22 (Humidity)

Results:

Mean Absolute Difference between runs: ± 0.48 %RH

Standard Deviation of Differences: 0.21 %RH

This shows that the DHT22 sensor gave consistent readings across repeated trials, with minimal variation, confirming good repeatability for humidity measurement.

Stability Analysis – DS18B20

In a stable section of the 16-hour atmospheric logging test, the DS18B20 sensor recorded temperature values with a mean of 27.12 °C and a standard deviation of 0.04 °C. The readings varied between 27.05 °C and 27.19 °C, with a total fluctuation range of 0.14 °C.

These results demonstrate very high stability and low noise, making DS18B20 suitable for long-term temperature monitoring.

Stability Analysis – DHT22

During the stable phase of the salt slurry experiment, the DHT22 sensor recorded humidity values with a mean of 75.97 %RH and a standard deviation of 0.07 %RH. The readings ranged from 75.8 %RH to 76.1 %RH, resulting in a total fluctuation of 0.3 %RH.

This indicates excellent stability and minimal sensor noise under constant humidity conditions.